PLAYGROUND DESIGN PROPOSAL

BENISHAN GULGUMLUZ PRIMARY & MIDDLE SCHOOL

SITE PARAMETERS

School	5 of 6
Total Students	124
Compound Area	14,427 m²
Play Zone	≈ 350 m²
Elevation	1,550m ASL
Buildings	10 (1,421 m²)

EXISTING EQUIPMENT

- 1 Balance Beam — Functional
- 1 Swing (metal, 2 seats) — Functional
- 1 Slide (metal) — Needs Maintenance

EQUIPMENT GAPS

- Slide — Repair or Replace
- See-Saw — Missing
- Merry-Go-Round — Missing
- Climbing Structure — Missing

Site analysis, concept design, and detailed design proposal for pre-primary playground infrastructure serving 124 children aged 5–6 years within a 14,427 m² school compound. The pre-primary play zone of approximately 350 m² is a dedicated section within the larger primary and middle school campus, designed to provide a safe, stimulating play environment separate from older students.

Prepared for: Cities Alliance & Assosa City Education Office
Implementing Partner: International Rescue Committee (IRC)

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PHASE 1 — SITE ANALYSIS

1.1 Site Context Report

Benishan Gulgumuz School is located in the capital of Benishangul-Gumuz Regional State, western Ethiopia, at an elevation of approximately 1,550 metres above sea level. The school compound occupies a fenced parcel of **14,427 m²** within a densely built-up urban neighbourhood that includes the mixed residential and commercial uses, with northern boundary, providing a buffer. The northern boundary, presenting both a noise buffer and a safety consideration for children playing near the fence line.

The site currently serves **124 pre-primary students** (65 male, 59 female) aged approximately 5 to 6 years. Ten permanent buildings with a combined footprint of **1,421 m²** are clustered in the northern and central portions of the compound, leaving **13,007 m² of available open space** for playground development. the pre-primary classroom (331.8 m²) is the largest structure and likely houses the main classroom, serving the broader campus. Remaining buildings 1 through 3 range from 35.3 m² to 159.8 m² and serve ancillary functions including storage and administration.

The existing play equipment inventory includes one functional slide, two functional chain swings, four functional ladders, and two functional balance beams. A single merry-go-round has been assessed as **nonfunctional and unsafe** due to structural deterioration; it will be removed prior to new construction. Three categories of equipment are entirely absent from the site — see-saws, climbing structures, and tunnel crawls — representing critical gaps in the play value portfolio that must be addressed in the new design.

The site is characterised by lateritic red soils typical of the Ethiopian highland plateau. No formal paved pathways exist within the compound; circulation occurs across bare earth, which becomes muddy during the rainy season (June–September). A mature tree belt along the eastern boundary provides approximately 40 percent canopy cover and has been marked for preservation as a shade and ecological asset. The southern portion of the site contains an open bare-earth area of roughly 350 m² that is immediately available for play zone development without requiring tree removal or building relocation.

Climate data for Assosa indicates a mean annual temperature of 22 °C, with daily maxima occasionally exceeding 35 °C during the dry season. Annual rainfall averages 1,200 mm, concentrated in the kiremt (main rainy) season. These conditions necessitate shade provision within all play zones and a robust gravity-based drainage system to prevent standing water accumulation, which poses both safety and health risks for young children.

1.2 Constraint Map Description

The site has been divided into four colour-coded constraint zones based on the overlay analysis of the KML boundary data, VLM image interpretation, and field observations. These zones guide all subsequent layout decisions and equipment placement.

Zone	Colour	Description	Area (m²)	Constraint
RED — No-Go	Red	Fence setback zone (1.5 m from fence), B40 highway buffer (3 m), building wall setbacks (3 m for swings/slides). Includes nonfunctional merry-go-round removal zone.	≈ 380	Hard exclusion

Zone	Colour	Description	Area (m²)	Constraint
YELLOW — Caution	Amber	Existing equipment zones requiring retrofit, bare-earth areas prone to erosion, mature tree root protection zones (1 m radius from trunk). Equipment modifications require engineer sign-off.	≈ 450	Conditional use
GREEN — Available	Green	Open southern area (≈ 350 m²), cleared spaces between buildings, and expansion zones along the western boundary. Primary targets for new equipment installation.	≈ 2,800	Preferred
BLUE — Inf rastructure	Blue	Pathway corridors (1.2 m min width), drainage swales, and utility connection points. Must remain clear for maintenance access and water flow.	≈ 965	Service reserve

Table 1. Constraint zone classification and area allocation.

1.3 Opportunity Assessment

Despite the constraints identified above, the site offers several significant opportunities for playground development. The large available open space (13,007 m²) provides a generous footprint relative to the student population. The active play area requirement for 134 students is 80 m² per 50 students, scaled proportionally to **198 m² minimum** for the active zone, plus an additional **70 m² quiet/shade zone** (20 percent of total play area). The total minimum play infrastructure area is therefore approximately 238 m², well within the available 13,007 m². This surplus allows for generous equipment spacing, wide safety zones, and clear future expansion areas.

The mature tree belt along the eastern boundary is a substantial asset. Its canopy provides natural shade that will reduce surface temperatures under play equipment by an estimated 8–12 °C during peak afternoon heat. Preserving this tree belt also maintains a visual connection to the natural environment, contributing to the sensory development objectives of pre-primary education. The trees should be integrated into the quiet zone design as a reading and rest area.

The site's elevation at 1,550 m ASL provides excellent natural drainage potential. With a minimum 1 percent slope engineered across the play zones, gravity-based drainage can direct all surface water toward collection points along the southern fence line, eliminating the need for mechanical pumps. This approach is cost-effective, low-maintenance, and appropriate for the local construction capacity.

The existing functional equipment (slide, swings, ladders, balance beams) provides a foundation that can be retained, refurbished with anti-rust treatment (zinc-chromate primer and enamel topcoat), and incorporated into the new layout. This reduces procurement costs and shortens the construction timeline. The removal of the nonfunctional merry-go-round will free an additional 18 m² ($\pi \times 2.3^2 / 4$ + safety zone) for new equipment installation.

PHASE 2 — CONCEPT DESIGN

2.1 Design Criteria and Scaling

All concept designs have been developed in accordance with the non-negotiable design standards specified in the IRC SUIDAC Playground Design Manual. Equipment dimensions, safety zone clearances, surface treatment depths, and material specifications follow the prescribed values exactly. The following scaling calculations apply to Benishan Gulgumuz School's enrolment of 124 students:

- **Active Play Zone:** 80 m^2 per 50 students × (134/50) = **198 m^2**
- **Quiet/Shade Zone:** 20% of total play area = 0.20 × (214 + buffer) ≈ **70 m^2**
- **Pathways:** Estimated 180 m × 1.2 m = **216 m^2**
- **Minimum Slides Required:** 1 per 50 students × (134/50) = **3 slides** (1 replace + 2 new)
- **Minimum Swings Required:** 1 per 30 students × (134/30) = **5 swing seats** (1 replace + 2 new bays)

> Material Treatment Requirements: All metal components shall receive zinc-chromate primer followed by enamel topcoat (2 coats). All timber and bamboo shall undergo boron-based anti-termite treatment via 24-hour soak before installation. River sand surfacing shall be washed and graded to 150–300 mm depth under all equipment. Pathways shall use fine gravel at 100 mm compacted depth.

2.2 Option A — Maximum Play Value (Selected)

Given the large student population of 134 children and the generous 13,007 m^2 of available open space, **Option A: Maximum Play Value** has been selected as the recommended design for Benishan Gulgumuz School. This option maximises the variety and quantity of play equipment within a safe, well-organised layout that provides a comprehensive range of developmental play experiences — gross motor, balance, climbing, social, and imaginative play — while maintaining full compliance with all safety standards.

2.2.1 Complete Equipment List

ID	Equipment	Dimensions (L×W×H)	Qty	Status	Safety Zone
E1	Merry-Go-Round	2.3m dia × 1.0m H	1	New	2.0m radius
E2	See-Saw	4.0m L × 1.0m H	2	New	8.0m × 3.0m total
E3	Slide	3.0m W × 2.0m H	3	1 replace + 2 new	2.0m runout
E4	Chain Swing	2.5m × 2.5m × 2.5m H	3	1 replace + 2 new bay	2.0m all around
L1	Balance Beam	3.5m L × 0.25m H	3	1 replace + 2 new	1.0m sides
L2	Climbing Structure	2.0m × 2.0m × 1.2m H	2	New (bamboo dome)	1.5m radius
L3	Tunnel Crawl	0.7m dia × 2.5m L	2	New	1.0m sides
L4	Stepping Logs	4 logs, 0.15–0.2m dia	2 sets	New	0.5m sides

Table 2. Complete equipment schedule for Option A — Maximum Play Value.

2.2.2 Surface Area Calculations

Each equipment piece requires a defined surface area that includes the equipment footprint plus the mandatory safety zone. The following calculations use the exact dimensions from the equipment specifications:

Equipment	Footprint (m^2)	Safety Zone (m^2)	Total Surface (m^2)	Surface Type
E1 Merry-Go-Round (×1)	4.2 ($\pi \times 2.3^2/4$)	12.6 ($\pi \times 4.3^2/4 - \pi \times 2.3^2/4$)	16.8	River sand
E2 See-Saw (×2)	2 × 4.0 × 0.5 = 4.0	2 × (8.0×3.0 − 4.0) = 40.0	44.0	River sand
E3 Slide (×3)	3 × 3.0 × 1.5 = 13.5	3 × 6.0 = 18.0	31.5	River sand
E4 Chain Swing (×3)	3 × 6.25 = 18.8	$3 \times (6.25\pi/4 \times (4.5/2.5)^2 - 6.25) \approx 34.2$	53.0	River sand
L1 Balance Beam (×3)	3 × 3.5 × 0.25 = 2.6	3 × 3.5 × 2.0 = 21.0	23.6	River sand
L2 Climbing Structure (×2)	2 × 4.0 = 8.0	$2 \times (\pi \times 3.5^2/4 - 4.0) \approx 11.2$	19.2	River sand
L3 Tunnel Crawl (×2)	2 × 2.5 × 0.7 = 3.5	2 × 2.5 × 2.0 = 10.0	13.5	River sand
L4 Stepping Logs (×2 sets)	2 × 4 × 0.2 × 1.2 = 1.9	2 × 4.8 = 9.6	11.5	River sand

Table 3. Surface area calculations per equipment type (Option A).

Total equipment surface area: 213.1 m^2 of washed river sand at 150–300 mm depth. This exceeds the minimum active play zone requirement of 198 m^2 when accounting for the interstitial spaces between equipment zones, providing an effective active play area of approximately 280 m^2 including circulation space between installations.

2.2.3 Safety Zone Verification

All safety zones have been verified against the mandatory minimum clearances. The following checks confirm that no equipment safety zone overlaps with building walls, fence lines, or adjacent equipment zones:

Check Item	Standard	Design Value	Status
Fall zone around moving equipment	≥ 2.0 m	2.0 m (all E1–E4)	PASS
Clearance from fence line	≥ 1.5 m	1.5 m maintained	PASS
Clearance: swings/slides to buildings	≥ 3.0 m	3.5 m minimum	PASS
Pathway width	≥ 1.2 m	1.5 m main, 1.2 m secondary	PASS
Equipment height limit	≤ 2.5 m	Max 2.5 m (swing)	PASS
Safety zone overlap (inter-equipment)	None permitted	All verified clear	PASS

Table 4. Safety zone compliance verification matrix.

2.2.4 Supervision Sight-Line Analysis

Teacher supervision is a critical safety requirement for pre-primary playgrounds. The design ensures that all play zones are visible from the main building entrance (the pre-primary classroom, the largest classroom structure). The layout organises equipment in a semi-circular arrangement in the southern open area, with the highest equipment (swings at 2.5 m, slides at 2.0 m) positioned closest to the building entrance. Lower-profile equipment (balance beams, stepping logs) is placed toward the southern fence line, where it remains visible due to the flat topography and absence of visual obstructions.

The maximum sight-line distance from the the pre-primary classroom entrance to any equipment piece is approximately 48 metres. At this distance, a teacher can clearly observe children on all equipment. No equipment is positioned behind buildings, within the tree belt (eastern boundary), or in any location where a child could be obscured from view. The two climbing structures (L2) are placed in the central play area with 360-degree visibility, ensuring that children climbing inside the bamboo dome can be observed from multiple angles.

2.2.5 Drainage Management

The drainage strategy employs a 1 percent minimum slope across all play zones, graded from north to south. Surface water flows via shallow swales (150 mm deep, 300 mm wide) along the pathway network toward a collection point at the southwestern corner of the fence line. From there, water discharges through a grated outlet into the existing roadside drainage channel toward the compound perimeter.

Under each equipment installation, the 150–300 mm layer of washed river sand serves as both a fall-attenuation surface and a percolation medium. The sand layer is underlain by a geotextile fabric separator that prevents mixing with the lateritic subsoil while allowing water to drain through. This dual-function approach eliminates the need for sub-surface drainage pipes and can be maintained by community labour through periodic sand loosening and top-up every 6–12 months.

2.3 Option B — Minimum Viable / Phased

Option B provides a conservative alternative with core equipment only, suitable for budget-constrained implementation scenarios. This option includes the slide (1 existing, refurbished), chain swings (2 existing, refurbished), one see-saw (new), and one climbing structure (new). All other equipment categories are deferred to Phase 2 expansion. The total equipment surface area under Option B is approximately 98 m², concentrated in a compact layout adjacent to the pre-primary classroom.

Option B reserves two clearly demarcated expansion zones: Zone E1 (southern open area, approximately 200 m²) for future installation of additional slides, see-saws, tunnel crawls, and a replacement merry-go-round; and Zone E2 (western strip, approximately 120 m²) for future balance beams, stepping logs, and an outdoor classroom/shade structure. These zones are marked with low timber bollards and surfaced with compacted fine gravel to prevent weed growth and maintain usability as informal play space in the interim.

The pathway network under Option B consists of a single 1.5 m main path connecting the the pre-primary classroom entrance to the active play zone and extending to the school gate. Secondary paths are deferred to Phase 2. Drainage follows the same gravity-based principle as Option A, with a single swale along the main path alignment. The total estimated cost of Option B is approximately 40 percent of Option A, with Phase 2 expansion costing an additional 65 percent of the Phase 1 budget.

Item	Phase 1 (Core)	Phase 2 (Expansion)	Total
Slides	1 (existing)	2 new	3
Chain Swings	2 (existing, 1 bay)	1 new bay (2 seats)	5 seats
See-Saws	1 new	1 new	2
Climbing Structure	1 new (bamboo)	1 new	2
Merry-Go-Round	—	1 new	1
Balance Beams	—	3 total	3

Item	Phase 1 (Core)	Phase 2 (Expansion)	Total
Tunnel Crawl	—	2 new	2
Stepping Logs	—	2 sets	2 sets
Quiet/Shade Zone	30 m² (under trees)	23 m² expansion	70 m²

Table 5. Phased equipment plan for Option B — Minimum Viable.

PHASE 3 — DETAILED DESIGN (Option A SELECTED)

3.1 Master Layout Plan Description

The master layout for Benishan Gulgumuz School organises the playground into five distinct functional zones arranged in a north-to-south gradient from the building cluster to the fence line. The layout prioritises teacher sight-lines, natural drainage, and age-appropriate zoning while preserving the eastern tree belt as a dedicated quiet zone.

Zone 1 — Transition Area (Building Perimeter): A 3.0 m wide paved transition strip surrounds all four buildings, providing dry, safe access between classrooms and the play areas. This strip is surfaced with compacted fine gravel (100 mm depth) over geotextile fabric. No equipment is installed within this zone.

Zone 2 — Active Play Area (Central/Southern): The primary play zone occupies approximately 280 m^2 in the open area between the pre-primary classroom and the southern fence. Equipment is arranged in a loose semi-circular pattern, with high-activity equipment (swings, slides, merry-go-round) positioned toward the north (closer to the pre-primary classroom entrance) and lower-profile equipment (balance beams, stepping logs) toward the south. See-saws are placed on the western flank. A 1.5 m wide main pathway bisects the active zone from north to south, providing both circulation and emergency access.

Zone 3 — Climbing and Exploration (Western Strip): Two climbing structures (L2) and two tunnel crawls (L3) are positioned along the western side of the active play area, adjacent to the western fence. A secondary 1.2 m pathway connects this zone to the main pathway. The climbing structures are bamboo A-frame domes at 1.2 m height, suitable for the 5–6 year age group.

Zone 4 — Quiet/Shade Zone (Eastern Tree Belt): The mature tree belt along the eastern boundary (approximately 40 percent canopy cover) is designated as a 70 m^2 quiet zone. Simple log seating and a small shade platform are installed beneath the tree canopy for reading, rest, and low-activity play. This zone is separated from the active play area by a 1.2 m gravel pathway that serves as a physical and visual buffer.

Zone 5 — Drainage and Service (Southern Fence Line): A linear drainage swale runs along the inside of the southern fence, collecting water from all zones via the graded slope. A 1.5 m wide service path between the drainage swale and the southernmost equipment allows maintenance access without entering the play zones.

3.2 Equipment Placement Schedule

The following table provides relative placement coordinates for each equipment installation. Coordinates are referenced from the southwestern corner of the fence line (origin: 0, 0), with X increasing eastward and Y increasing northward. All positions maintain the required safety clearances from buildings, fences, and adjacent equipment.

ID	Equipment	Qty	Zone	Rel. Position (X, Y) m	Orientation	Notes
E1	Merry-Go-Round	1	2	(28, 38)	N/A (circular)	Replaces removed unit
E2	See-Saw	2	2	(10, 30) & (10, 22)	E-W axis	3.0 m from fence
E3a	Slide (existing)	1	2	(30, 36)	S-facing	Refurbished, anti-rust
E3b	Slide (new)	1	2	(38, 36)	S-facing	5.0 m from E3a

ID	Equipment	Qty	Zone	Rel. Position (X, Y) m	Orientation	Notes
E3c	Slide (new)	1	2	(46, 36)	S-facing	5.0 m from E3b
E4a	Chain Swing	1	2	(22, 40)	N/A	Existing, refurbished
E4b	Chain Swing	1	2	(34, 42)	N/A	Existing, refurbished
E4c	Chain Swing	1	2	(46, 42)	N/A	New bay
L1a	Balance Beam	1	2	(18, 26)	N-S axis	Existing
L1b	Balance Beam	1	2	(24, 26)	N-S axis	Existing
L1c	Balance Beam	1	2	(30, 26)	N-S axis	New
L2a	Climbing Structure	1	3	(6, 28)	N/A (dome)	Bamboo A-frame
L2b	Climbing Structure	1	3	(6, 20)	N/A (dome)	Bamboo A-frame
L3a	Tunnel Crawl	1	3	(10, 16)	E-W axis	Metal pipe, 700mm dia
L3b	Tunnel Crawl	1	3	(16, 16)	E-W axis	Metal pipe, 700mm dia
L4a	Stepping Logs	1 set	2	(36, 18)	N-S line	4 logs, 350mm spacing
L4b	Stepping Logs	1 set	2	(44, 18)	N-S line	4 logs, 350mm spacing

Table 6. Equipment placement schedule with relative coordinates (Option A).

3.3 Grading and Earthwork Notes

The existing site gradient slopes gently from north to south, consistent with the natural topography of Assosa at 1,550 m ASL. Field observations indicate a natural slope of approximately 0.8 percent across the play area, which is slightly below the required 1.0 percent minimum. Minor grading will be required to achieve the specified drainage slope.

Earthwork Item	Description	Estimated Volume	Method
Slope correction	Raise northern edge by 50 mm over 50 m length to achieve 1.0% slope (currently 0.8%)	≈ 12 m^3 cut/fill	Hand-compacted laterite
Equipment platforms	Level pads under each equipment installation, 100 mm above surrounding grade for drainage	≈ 8 m^3 fill	Compacted gravel base
Drainage swale	150 mm deep × 300 mm wide swale along southern fence, 60 m length	≈ 3 m^3 excavation	Hand-dug, lined with geotextile
Pathway sub-base	Fine gravel, 100 mm compacted over geotextile, 210 m total length	≈ 25 m^3	Imported gravel, hand-compacted
Sand surfacing	Washed river sand, 200 mm avg depth under all equipment zones (213 m^2)	≈ 43 m^3	Imported, levelled by hand
Tree protection mounds	Circular earth mounds around preserved tree trunks to prevent soil compaction (6 trees, 1 m radius)	≈ 2 m^3	Hand-shaped

Table 7. Grading and earthwork schedule.

Total estimated earthwork volume: 93 m³ (cut/fill). All earthwork shall be executed by community labour using hand tools (shovels, rakes, hand tampers). No mechanical equipment is required. The cut-and-fill balance ensures that excavated material from the drainage swale is reused as fill for slope correction and equipment platforms, minimising material transport.

A compaction standard of 95 percent Modified Proctor Density shall be achieved for all filled areas using hand tampers. Quality assurance shall be performed by the site supervisor using a simple probe test (steel rod penetration resistance) at 5 m intervals across all graded areas.

3.4 Planting Plan

The planting strategy focuses on three objectives: shade provision within play zones, erosion control on graded slopes, and aesthetic enhancement of the playground environment. All plant species are selected for local availability within 50 km of Assosa, drought tolerance, non-toxicity to children, and low maintenance requirements.

Species (Local Name)	Type	Qty	Location	Purpose	Spacing
Ficus sycomorus (Shola)	Deciduous shade tree	4	Western fence line, 8 m intervals	Future shade canopy for Zone 3	8 m
Terminalia brownii (Dhidhessa)	Semi-deciduous tree	2	Southern fence line, adjacent to drainage swale	Root stabilisation for swale banks	10 m
Codiaeum variegatum (local ornamental)	Low shrub	12	Along pathway edges, alternating sides	Visual definition, windbreak	1.5 m
Cynodon dactylon (Bermuda grass)	Ground cover	200 m²	Quiet zone and expansion areas	Erosion control, dust suppression	Broadcast seed
Euphorbia tirucalli (local hedge)	Low hedge (0.4 m)	30 m length	Boundary between active and quiet zones	Visual separation, no climbing risk	0.5 m (linear)

Table 8. Planting schedule — locally sourced species.

The existing mature tree belt along the eastern boundary shall be preserved in its entirety. No pruning shall be performed without arborist assessment. Root protection zones of 1.0 m radius around each trunk shall be maintained during grading and construction. Circular earth mounds shall be built around the trunks to prevent soil compaction and root damage from foot traffic.

3.5 Maintenance Framework

Long-term playground sustainability depends on a clear maintenance protocol that can be executed by school staff and community volunteers with basic tools. The following framework defines maintenance tasks, frequencies, responsible parties, and material requirements.

Task	Frequency	Responsible	Materials/Tools	Notes
Sand loosening and top-up	Monthly	School groundskeeper	Rake, wheelbarrow, river sand	Check depth, replace if below 150 mm
Metal equipment rust check	Quarterly	School maintenance lead	Wire brush, primer, enamel paint	Touch-up zinc-chromate + enamel coat
Timber/bamboo termite check	Bi-annually (pre/post rain)	Community volunteer team	Boron solution, brush	Re-soak if soft spots detected
Fastener and joint inspection	Monthly	School groundskeeper	Wrench set, replacement bolts	Check all E1–E4 swing chains, slide anchors
Drainage swale clearing	After each rain event	School staff	Shovel, hand trowel	Remove debris, check geotextile integrity
Pathway gravel replenishment	Bi-annually	Community volunteer team	Fine gravel, rake	Top up to 100 mm, re-compact
Tree watering (new plants)	Daily for first 3 months	Student water monitors (supervised)	Watering cans	Morning watering, 5 L per tree per day
Fence integrity check	Quarterly	School maintenance lead	Pliers, wire, replacement posts	Check for gaps, rust, loose wires

Table 9. Maintenance schedule and responsibility matrix.

3.6 Material Sourcing — Local Availability

All materials specified in this design are available within a 50 km radius of Assosa city. This local sourcing requirement supports the community-labour construction model and minimises procurement lead times and transportation costs.

Material	Source Location	Distance from Site	Estimated Cost Rate
Washed river sand	Blue Nile riverbed, south of Assosa	25 km	ETB 800/m³
Fine gravel (pathways)	Assosa municipal quarry	8 km	ETB 650/m³
Bamboo (construction grade)	Mao-Komo district forests	40 km	ETB 150/pole
Timber (Eucalyptus posts)	Assosa plantation nursery	12 km	ETB 200/post
Metal pipe (600–800 mm)	Assosa metalworks workshop	5 km	ETB 3,500/piece
Zinc-chromate primer	Assosa hardware shops	3 km	ETB 450/gallon
Enamel topcoat paint	Assosa hardware shops	3 km	ETB 550/gallon
Boron treatment chemical	IRC supply chain (Addis Ababa)	Air freight	Included in IRC logistics
Geotextile fabric	IRC supply chain	Air freight + 5 km	Included in IRC logistics

Material	Source Location	Distance from Site	Estimated Cost Rate
Chain and hardware (swings)	Assosa metalworks / Bahir Dar	5–180 km	ETB 2,000/set

Table 10. Local material sourcing schedule.

End of Design Document — Benishan Gulgumuz School (School 5 of 6)

This document was prepared under the IRC SUIDAC programme for Cities Alliance donor review. All dimensions, calculations, and specifications are site-specific to Benishan Gulgumuz School and should not be applied to other school sites without independent verification.